

Next-Token Prediction

Tokens, embeddings and neural language models

Prof. Nipun Batra

ES 667 · Deep Learning · IIT Gandhinagar

What is sequence modeling?

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This lecture changes the **input type**. We move from fixed tensors to **sequences** of variable length — the first step toward language models.

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audio	waveform samples / frames
sensors	readings over time
events / logs	user clicks, actions
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length T **varies** from example to example — a fixed-size vector no longer fits

Order carries meaning

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A model f over sequences must **preserve order**. Treating $x_{1:T}$ as an unordered bag throws away exactly the structure that makes it a sequence.

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classification $x_{1:T} \rightarrow y$



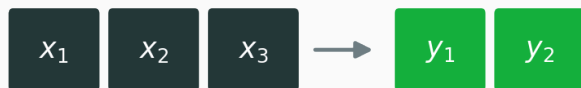
single label

token classification $\rightarrow y_{1:T}$



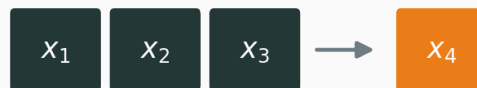
one label / token

seq2seq $\rightarrow y_{1:U}$



new length U

next-token $x_{1:t} \rightarrow x_{t+1}$



predict the next

The sequence tasks, precisely

Task	Shape	Example
classification	$x_{1:T} \rightarrow y$	sentiment of a review
token classification	$x_{1:T} \rightarrow y_{1:T}$	tag each word (POS)
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this lecture: **next-token prediction** — the task behind language models

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not a single guess — a **score for every possible next token**

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This is Karpathy's **makemore** (part 2), which is exactly the neural language model of **Bengio et al. 2003, "A Neural Probabilistic Language Model"**. We build up to that model in this lecture.

The probability of a sequence

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How can one model score a sequence of **arbitrary length**? Factor it into pieces.

The chain rule of probability

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exact — no approximation. This holds for **any** sequence distribution.

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each step: a V -way classification of “which token comes next?”

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this is exactly **token-level cross-entropy** — the same loss as classification

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. (start)	e
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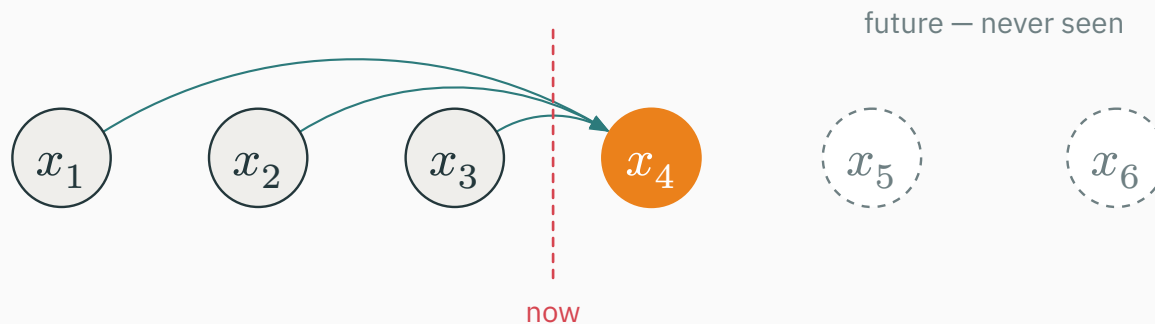
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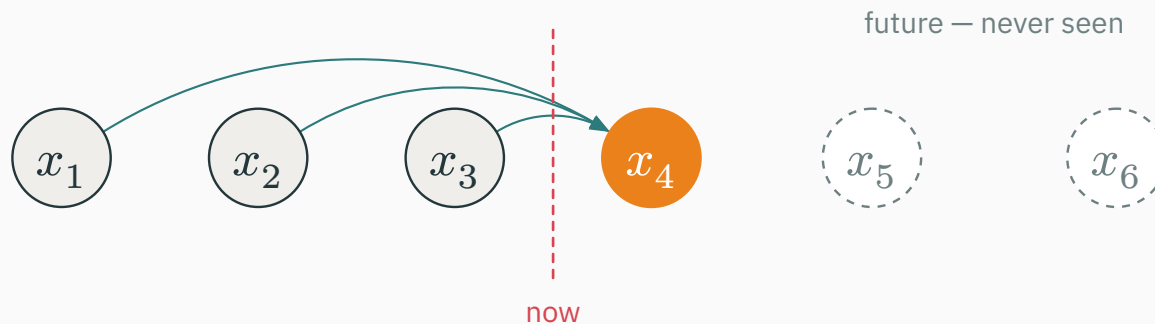
one 4-letter name $\rightarrow$ 5 training pairs. A corpus gives **millions**.

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autoregressive: each token is predicted from the tokens before it

Tokens and vocabulary

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makemore: $V = \{ ., a, b, \dots, z \}$ – 27 symbols

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Subword tokenization (e.g. **BPE**) is the modern default — a middle ground. We won't deep-dive it here; makemore just uses **characters**.

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makemore reuses a single . as both start and end marker

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we need a representation with **no spurious arithmetic — next**

One-hot vectors and embeddings

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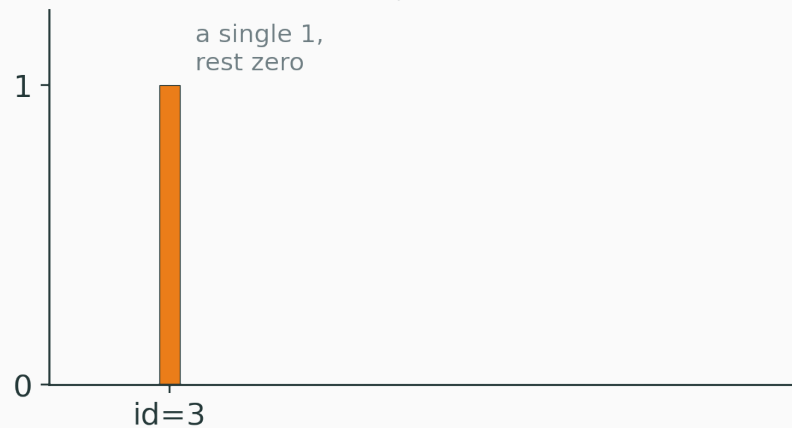
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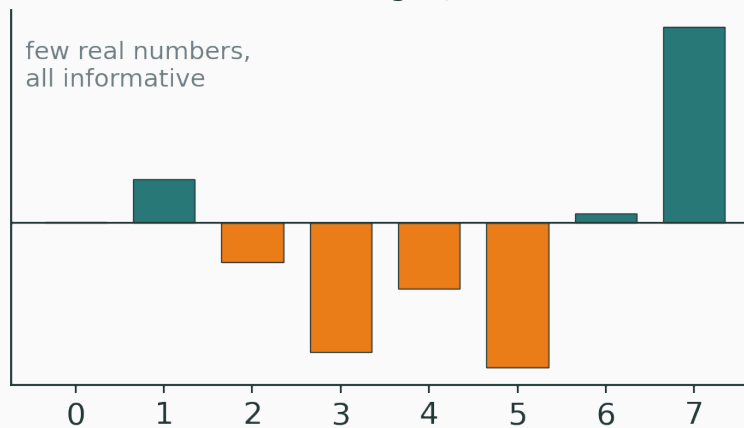
$$\|o_i - o_j\|^2 = 2 \text{ for } \textbf{every} \text{ distinct pair} - \text{“cat” is as close to “dog” as to “the”}$$

One-hot vs dense embedding

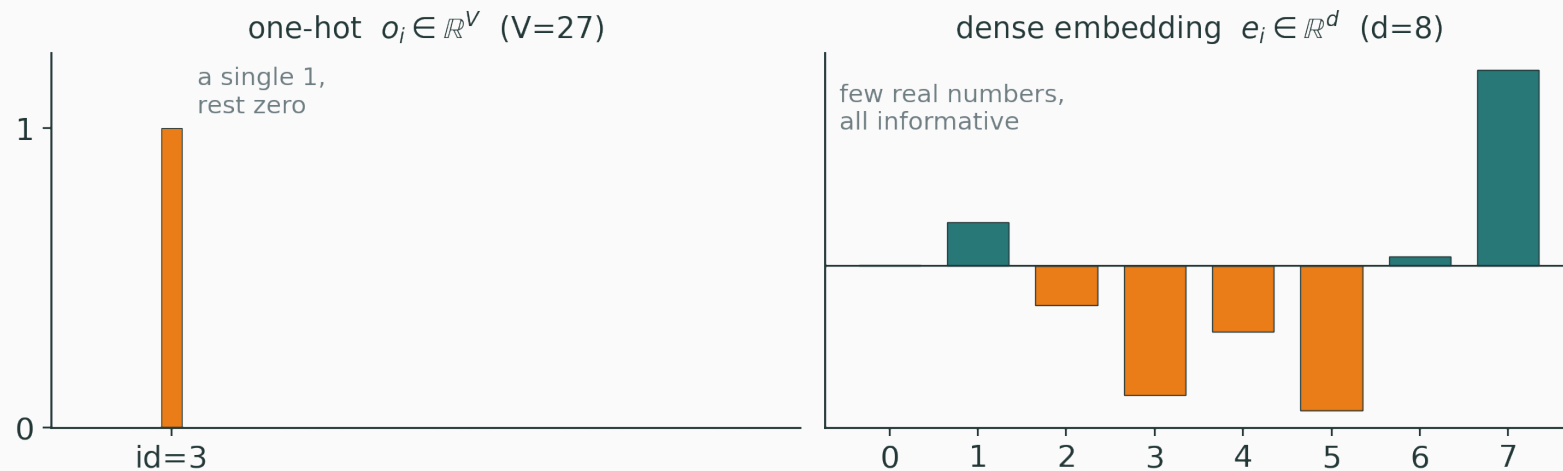
one-hot $o_i \in \mathbb{R}^V$ ($V=27$)



dense embedding $e_i \in \mathbb{R}^d$ ($d=8$)



One-hot vs dense embedding



replace a long sparse indicator with a short **dense** vector of learned features

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from a V -dim sparse axis to a d -dim dense point — d is the **embedding dimension**

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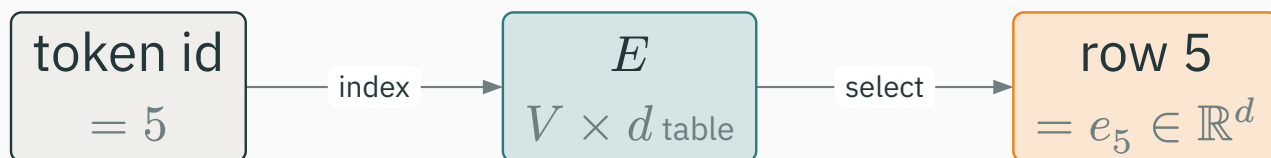
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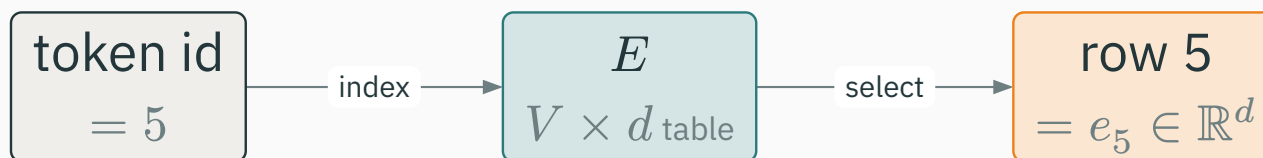
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in code it is a **table lookup** — but mathematically it is linear, so gradients flow through it

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rows 3 and 4: $(e_{\text{cats}}, e_{\text{dogs}}) = ((1, 1), (-1, 1))$ — two dense rows out, the whole embedding layer

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$E \in \mathbb{R}^{V \times d}$ (rows = tokens, cols = features)

token	.	+0.3	+0.8	+0.3	-1.3	
	e	+0.9	+0.4	-0.5	+0.6	$\leftarrow e_e$
	m	+0.4	+0.3	+0.0	+0.5	
	a	-0.7	-0.2	-0.5	+0.6	
	o	+0.0	-0.3	-0.8	-0.3	$\leftarrow e_o$
	l	+0.0	-0.3	+1.3	+1.0	
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looking up **e** and **o** picks out two rows of E — nothing else is touched

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$$E \leftarrow E - \eta \nabla_E \mathcal{L}$$

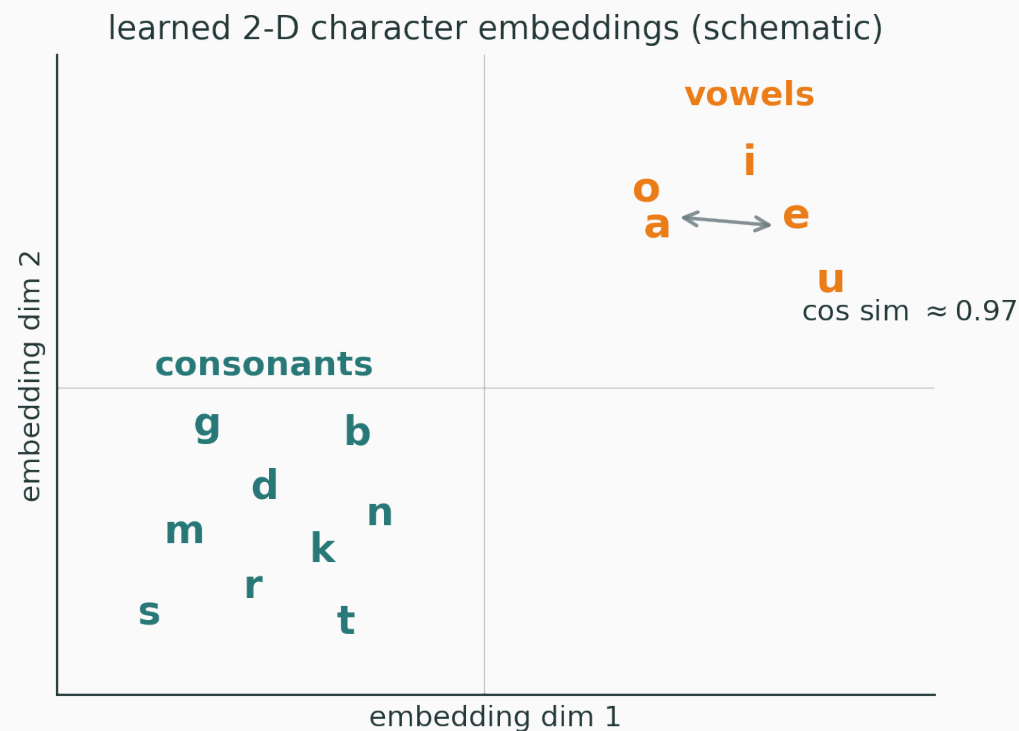
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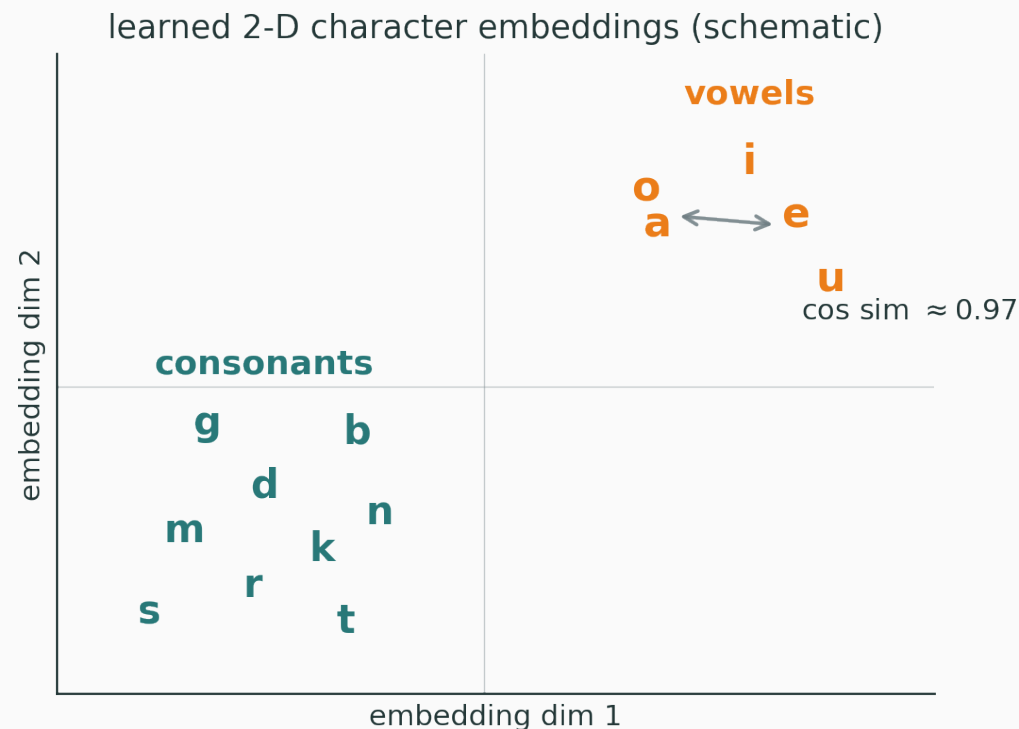
Only the **rows that were looked up** in a batch receive a gradient — an unseen token's embedding does not move on that step. Frequent tokens get trained the most.

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in makemore, vowels and consonants tend to cluster — measured by **cosine similarity**

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Do **not** over-promise. Not every direction is a clean semantic axis, and small models learn only coarse structure. “Vowels cluster” is a **tendency**, not a law.

INTERACTIVE

↗ nipunbatra.github.io/interactive-articles/embedding-lookup

click a token id and watch the corresponding **row** of E light up and drop out as the dense embedding vector; nudge a value and see the point move in 2-D.

A fixed-context MLP language model

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Simplify: condition on only the previous k tokens (a **fixed context window**):

$$p_{\theta}(x_t \mid x_{<t}) \approx p_{\theta}(x_t \mid x_{t-k:t-1})$$

The full history is too much

Exactly, $p(x_t \mid x_{<t})$ conditions on **all** previous tokens — an unbounded input.
Simplify: condition on only the previous k tokens (a **fixed context window**):

$$p_{\theta}(x_t \mid x_{<t}) \approx p_{\theta}(x_t \mid x_{t-k:t-1})$$

a bounded, fixed-size input $\rightarrow$ an ordinary MLP can handle it. This is the Bengio-2003 / makemore model.

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e m m	a
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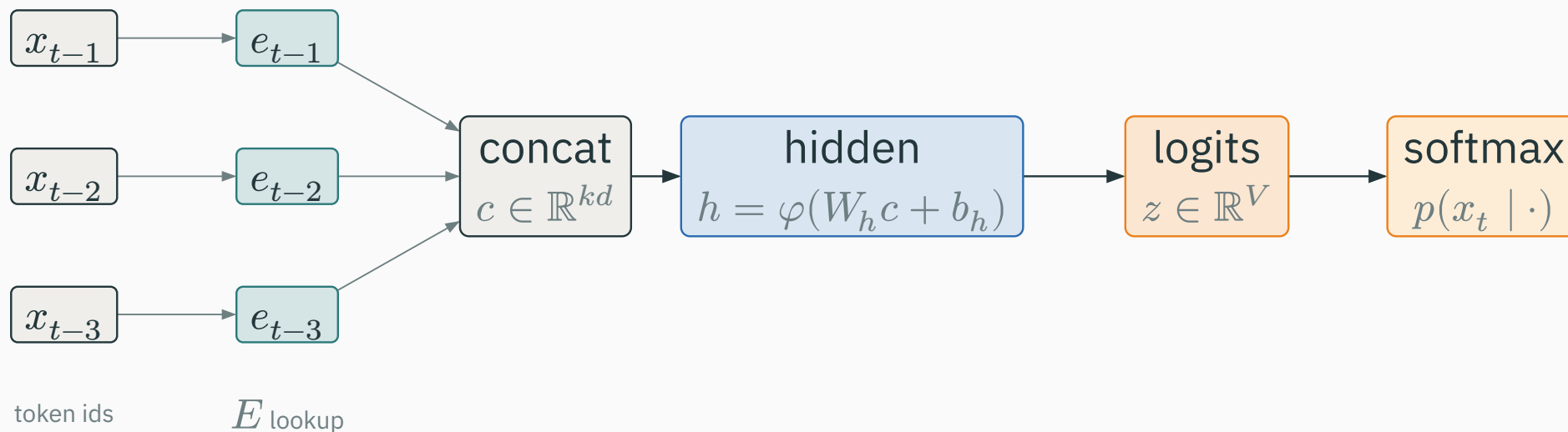
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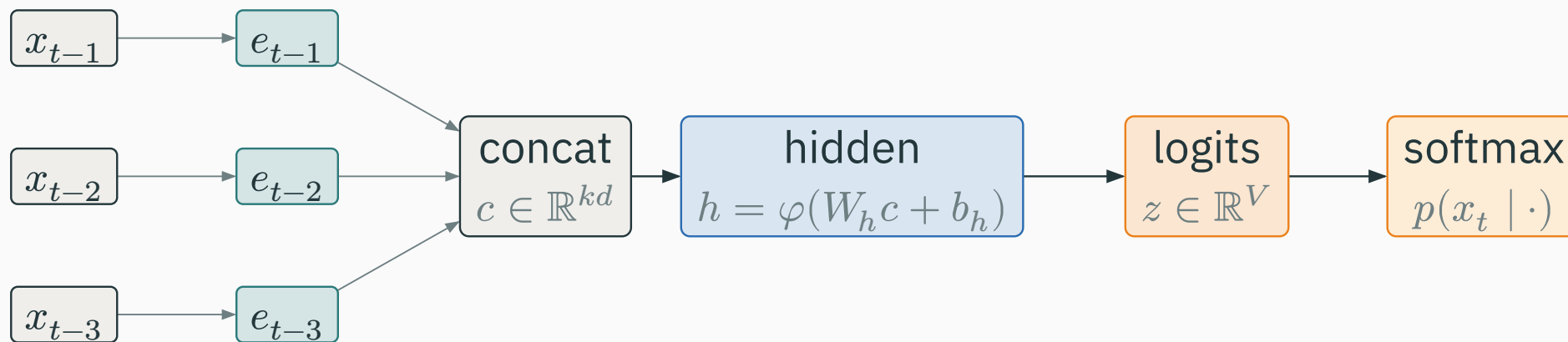
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pad the start with .; every window is one supervised example

The MLP language model



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token ids

E lookup

look up $\rightarrow$ concatenate $\rightarrow$ hidden layer $\rightarrow$ logits $\rightarrow$ softmax over V

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Bengio et al. 2003 · Karpathy's makemore — embeddings + one hidden layer + softmax

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each position lands in its **own** slice of c_t , hitting its **own** columns of W_h

Tensor shapes

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Tensor	Shape	Meaning
context ids X	$B \times k$	k token ids per row
embeddings $E[X]$	$B \times k \times d$	one d -vector per token
concat C	$B \times kd$	flatten the k embeddings
hidden H	$B \times H_{\text{dim}}$	after W_h and φ
logits Z	$B \times V$	one score per vocab token

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note: the **embedding** and **output** blocks both scale with the vocabulary V

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embedding + output (both $\propto V$) dominate — halving V nearly **halves** the model

A fully-worked forward and backward pass

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small enough to do **entirely by hand** — and check against the earlier formulas

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the first hidden unit fires; the second is switched off

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a bigger logit becomes an **exponentially** bigger weight — that is the “soft-max”

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the model already puts most mass (0.657) on the correct token cats

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a confident-but-imperfect prediction; a perfect one would give $\mathcal{L} = 0$

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push down the three wrong logits, **push up** the correct one `cats` — that is all learning is here

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the two **zeros** are the coordinates the switched-off hidden unit never touched

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only rows I and like — cats and dogs were not in the context, so their gradient is 0

INTERACTIVE

↗ nipunbatra.github.io/interactive-articles/makemore-forward

step through this exact example: watch the embeddings look up, the concat form, the softmax bar chart appear, and the $p - y$ gradient flow back and nudge the two used rows.

Training objective and evaluation

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the token-level cross-entropy from Part II — now over the whole sliding-window dataset

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the mask makes padding **invisible** to the gradient — only real tokens count

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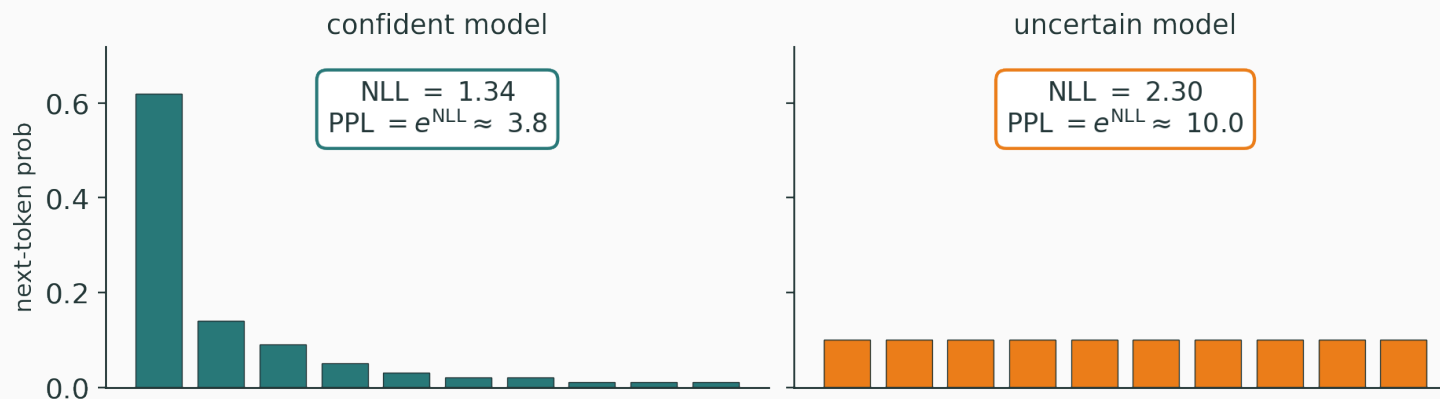
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Interpretation: the **effective number of plausible next tokens**. $\text{PPL} = 1$ means perfectly certain; $\text{PPL} = V$ means no better than a uniform guess.

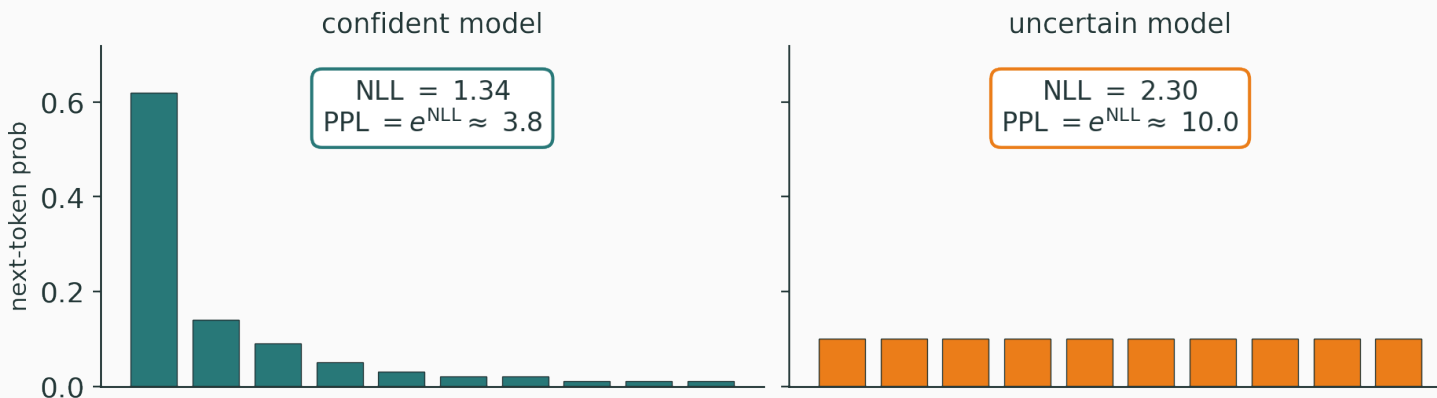
Perplexity, two ways

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a peaked distribution has **low** perplexity; a flat one is **perplexed** among many tokens

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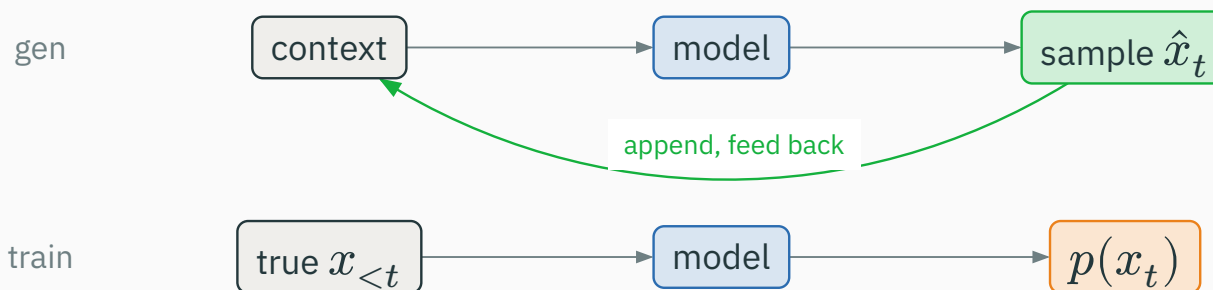
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Caveat. Perplexity depends on the **tokenization**. A character model and a word model are **not** comparable by PPL — they predict different-sized units.

Autoregressive generation

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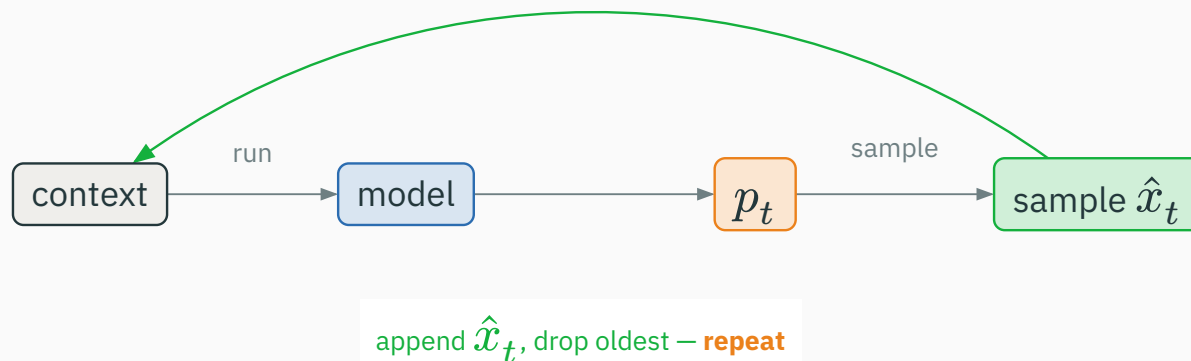
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Pro — deterministic, simple, always picks the model's best guess.

Con — repetitive and bland; locally-best $\neq$ globally-best sequence.

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. -> e -> m -> m -> a -> . yields emma; a different draw yields elian, amara, ...

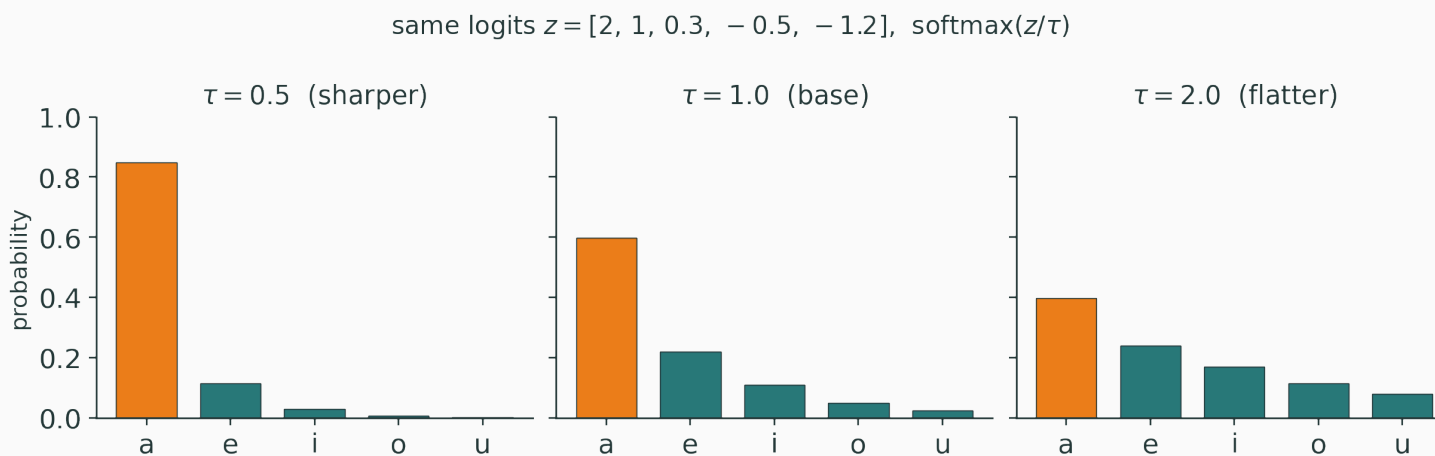
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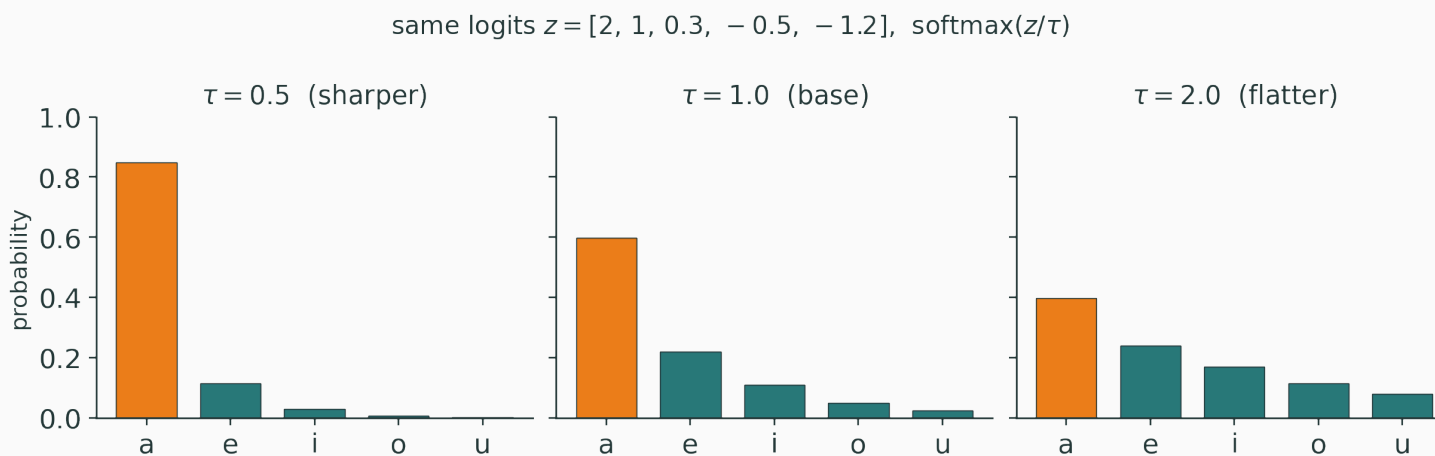
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$\tau < 1$ sharpens (bolder) · $\tau > 1$ flattens (more random) · $\tau \rightarrow 0 \rightarrow$ greedy

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nucleus / top- p sampling is the close cousin — we meet it in the Transformer lecture

INTERACTIVE

↗ nipunbatra.github.io/interactive-articles/softmax-temperature

Sweep temperature and top- k on a fixed set of logits and watch the next-token bar chart sharpen or flatten — then sample repeatedly and see the diversity of outputs change.

Where the fixed-context MLP fails

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To choose **was** vs **were** the model needs the subject **book** — but with a small window that word has already scrolled out of view. Long-range dependencies are lost.

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you cannot simply set $k = \infty$ — cost grows with the window, and k is fixed at build time

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The **same** token at position 1 vs position 2 is processed by **different** weights. Order is preserved, but computation is **not shared** across time — the model relearns each position separately.

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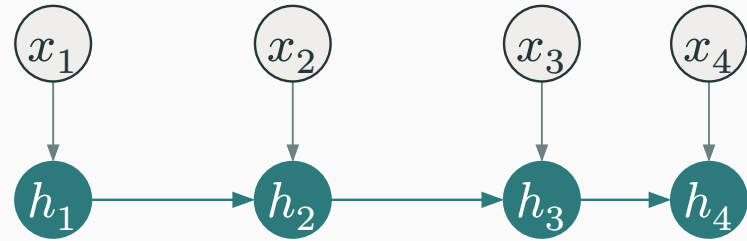
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a recurrent update: $h_t = f_{\theta}(h_{t-1}, x_t)$

From fixed window to recurrent state



fixed window (MLP)



state carries the past

recurrent (RNN)

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The hidden state h_t is a **learned summary** of everything seen so far — one shared function f_{θ} applied at every step, for any length.

One-slide summary

Idea	Takeaway
chain rule	$p(x_{1:T}) = \prod_t p(x_t \mid x_{<t})$ — model every conditional
language modeling	repeated next-token classification; loss = token cross-entropy
tokens	text $\rightarrow$ integer ids; ids are categorical, not numeric
embeddings	learned dense rows of E ; similar tokens land nearby
MLP-LM	fixed window $\rightarrow$ concat $\rightarrow$ MLP $\rightarrow$ softmax (Bengio / makemore)
generation	autoregressive; greedy / sample / temperature / top- k
limitation	fixed window, position-specific weights $\rightarrow$ motivates RNNs

Final mental model — one idea

A language model estimates a **distribution over the next token.**

Language modeling is **repeated classification**

tokens → embeddings → context → softmax over the vocabulary

We predicted the next token from a **fixed window**.

Next: **Sequence Models — RNNs** — hidden state, backprop through time, and LSTMs / GRUs that carry information across a whole sequence.